CORE PURE / PROOF BY INDUCTION

Proof by Induction - Matrices

Mark scheme 8 questions · 55 marks

Questions

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	Total	**55**	

Reading the mark scheme

These rules apply throughout, unless a note says otherwise.

Marks

Every mark is worth one, so **M1 A1** on a row is two marks. Only marks coded ft follow through.

Code	Meaning
M1	Method: a correct method, applied to this question. A slip doesn't cost it if the method is clear. A formula must be used, not just quoted.
dM1	Dependent method: as M1, but only if the method mark before it was earned, unless its note names another.
A1	Accuracy: a correct answer or step. It needs the method mark it follows.
A1ft	As A1, following through from an earlier answer that the note names.
A1 AG	The last mark of a part whose answer is printed in the question: complete working, with no errors.
B1	A correct value, fact or statement. It needs no method mark.
B1ft	As B1, following through from an earlier answer that the note names.
E1	A written explanation, interpretation or conclusion. Concluding a proof by induction needs every earlier mark.

Abbreviations

ft and AG qualify a code in the Marks column. The others appear in the notes.

ft	follow through	**AG**	answer given in the question
cao	correct answer only, with no follow-through	**cso**	correct solution only: no errors in the part
oe	or an equivalent form	**awrt**	any answer that rounds to this value
soi	seen, or implied by later working	**isw**	ignore working after a correct answer
SC	special case: credit outside the main method	**their**	the student's own earlier value
condone	a slip that doesn't cost the mark	[ ]	working the mark doesn't need
s.f.	significant figures	**d.p.**	decimal places

Reading the scheme

- Each row is a line of working, with the marks it earns beside it. Where a row carries two marks, each note starts with its code.
- [$\boldsymbol{n}$] ends each part with its total.
- A shaded *Alternative* is a second method for the same total. Mark an answer by the method it follows.
- Any other valid method earns the equivalent marks, unless the question names a method or says *hence*.
- A sketch earns one B1 for each feature named below its figure.

Accuracy and answers

- Unless a question says otherwise, give answers to 3 significant figures, and angles in degrees to 1 decimal place. Angles in radians take 3 significant figures.
- A final answer with more figures is accepted if it rounds to the scheme's value; one with fewer is not. An answer spoiled by rounding too early loses its accuracy mark.
- Where a question asks for an exact answer, a decimal loses the final accuracy mark. Otherwise either is accepted.
- Wrong or missing units cost nothing unless the question asks for them.
- An omitted $\mathrm{d}\theta$, $\mathrm{d}x$ or $\mathrm{d}u$ in an integral is not penalised where the variable of integration is clear from the integrand and the limits. It is penalised where it is genuinely ambiguous: a substitution part-way through, two variables in play, or limits in the wrong variable.
- A correct answer with no working earns full marks, unless the question needs the working: show that, prove, hence, a method it names, or an exact answer. A wrong answer with no working earns nothing. Calculator answers are treated the same way.
- A correct answer, once reached, isn't spoiled by later working (isw).

Value of $\boldsymbol{g}$

- Take $g = 9.8\,\mathrm{m\,s^{-2}}$ unless a question states a value, which is then the only one accepted. Answers that use $g = 9.8$ may be given to 2 or 3 significant figures.
- Where a question doesn't state g, an answer worked with $g = 10$ throughout earns the same marks, keeping any value printed in the question. The worked solutions list the final answers with $g = 10$.

Question	Scheme	Marks	Notes
1	$n = 1$: $\begin{pmatrix} 2^1 & 3^1 - 2^1 \\ 0 & 3^1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$	**B1**	Needs the formula evaluated at $n = 1$ and shown equal to the given matrix; 'true for $n = 1$' alone is not enough.
	Assume true for $n = k$: $\begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}^{k+1}$ $= \begin{pmatrix} 2^k & 3^k - 2^k \\ 0 & 3^k \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$	**M1**	States the assumption for $n = k$ and writes the $(k+1)$th power as the assumed kth-power matrix times the given matrix, in either order.
	$= \begin{pmatrix} 2^{k+1} & 2^k + 3(3^k - 2^k) \\ 0 & 3^{k+1} \end{pmatrix}$	**M1 A1**	**M1** Multiplies their matrices in either order, forming the top-right entry row by column. Condone slips. **A1** Correct product in either order; entries need not be simplified.
	$2^k + 3^{k+1} - 3 \times 2^k = 3^{k+1} - 2^{k+1}$	**A1**	Top-right entry correctly simplified to $3^{k+1} - 2^{k+1}$, other entries 2^{k+1}, 0, 3^{k+1}. Not for the result asserted directly from the product.
	True for $n = 1$; true for $n = k$ $\Rightarrow$ true for $n = k + 1$; so true for all $n \in \mathbb{Z}^+$	**E1**	Concludes with all three points: true for $n = 1$, true for $n = k$ implies true for $n = k + 1$, hence true for all positive integers.
	Total for Question 1	**6**	

Question	Scheme	Marks	Notes
2	$n = 1$: LHS $= \begin{pmatrix} 3 & -1 \\ 1 & 1 \end{pmatrix}$ RHS $= 2^0 \begin{pmatrix} 3 & -1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 3 & -1 \\ 1 & 1 \end{pmatrix}$	**B1**	Both sides evaluated at $n = 1$ and shown equal; a bare statement that it is true scores B0.
	Assumes $\mathbf{A}^k = 2^{k-1} \begin{pmatrix} k+2 & -k \\ k & 2-k \end{pmatrix}$ $\mathbf{A}^{k+1} = 2^{k-1} \begin{pmatrix} k+2 & -k \\ k & 2-k \end{pmatrix} \begin{pmatrix} 3 & -1 \\ 1 & 1 \end{pmatrix}$	**M1**	Assumes the result for $n = k$ and writes $\mathbf{A}^{k+1}$ as their $\mathbf{A}^k$ times $\mathbf{A}$, in either order.
	$= 2^{k-1} \begin{pmatrix} 3(k+2)-k & -(k+2)-k \\ 3k+2-k & -k+2-k \end{pmatrix}$	**M1 A1**	**M1** Multiplies the matrices in the order set up, attempting all four elements. Condone one slip. **A1** Correct product for their order, need not be simplified.
	$= 2^k \begin{pmatrix} k+3 & -(k+1) \\ k+1 & 2-(k+1) \end{pmatrix}$	**A1**	cso. Takes out the factor 2 to reach the result with $n = k + 1$, shown in that form or stated as the $n = k + 1$ result.
	True for $n = 1$; true for $n = k \Rightarrow$ true for $n = k + 1$ $\Rightarrow$ true for all $n \in \mathbb{N}$	**E1**	States true for $n = 1$, true for $n = k$ implies true for $n = k + 1$, and hence true for all $n \in \mathbb{N}$.
	Total for Question 2	**6**	

Question	Scheme	Marks	Notes
3(a)	$n=1$: $2^1\begin{pmatrix}2&1\\-1&0\end{pmatrix}=\begin{pmatrix}4&2\\-2&0\end{pmatrix}=\mathbf{A}$	**B1**	Formula evaluated at $n=1$ and shown equal to $\mathbf{A}$; a bare statement that it is true for $n=1$ scores nothing.
	Assume $\mathbf{A}^k=2^k\begin{pmatrix}k+1&k\\-k&1-k\end{pmatrix}$ $\mathbf{A}^{k+1}=2^k\begin{pmatrix}k+1&k\\-k&1-k\end{pmatrix}\begin{pmatrix}4&2\\-2&0\end{pmatrix}$	**M1**	States the assumption for $n=k$ and writes $\mathbf{A}^{k+1}$ as $\mathbf{A}^k\mathbf{A}$ or $\mathbf{A}\mathbf{A}^k$ with the assumed form substituted.
	$=2^k\begin{pmatrix}4(k+1)-2k&2(k+1)\\-4k-2(1-k)&-2k\end{pmatrix}$	**M1 A1**	**M1** Multiplies row by column for all four elements, in either order. Condone arithmetic slips. **A1** Correct product in either order; need not be simplified.
	$=2^{k+1}\begin{pmatrix}k+2&k+1\\-(k+1)&1-(k+1)\end{pmatrix}$	**A1**	Factor 2^{k+1} with every element shown equal to the formula's with $n=k+1$, by rewriting or by comparison with a stated target.
	True for $n=1$; true for $n=k\Rightarrow$ true for $n=k+1$ $\Rightarrow$ true for all positive integers n	**E1**	States true for $n=1$, true for $n=k$ implies true for $n=k+1$, hence true for all positive integers n.
		[6]	
3(b)	$\mathbf{A}^n=2^n\mathbf{M}$, $\mathbf{M}=\begin{pmatrix}n+1&n\\-n&1-n\end{pmatrix}$ $\det\mathbf{M}=(n+1)(1-n)+n^2=1$ $\mathbf{B}=2^{-n}\mathbf{M}^{-1}$	**M1**	Takes out 2^n, or works with $\mathbf{A}^n$: swaps the leading diagonal, negates the other two elements and divides by their determinant. Condone determinant slips.
	$\mathbf{B}=\frac{1}{2^n}\begin{pmatrix}1-n&-n\\n&n+1\end{pmatrix}$	**A1**	oe, such as 2^{-n} or the factor multiplied into each element.
		[2]	
	Total for Question 3	**8**	

Question	Scheme	Marks	Notes
4	$n = 1$: $2^1 = 2$, $1 \times 2^0 = 1$, $\frac{1(0)}{2}2^{-1} = 0$ $\Rightarrow$ RHS $= \mathbf{A}$	**B1**	RHS at $n = 1$ evaluated and shown equal to $\mathbf{A}$; not for stating "true for $n = 1$" alone.
	Assume true for $n = k$: $\mathbf{A}^k =$ the given matrix with $n = k$ $\mathbf{A}^{k+1} = \mathbf{A}^k\mathbf{A}$	**M1**	States the assumption for $n = k$ and writes $\mathbf{A}^{k+1}$ as $\mathbf{A}^k\mathbf{A}$ or $\mathbf{A}\mathbf{A}^k$ with the assumed matrix.
	Diagonal: $2^k \times 2 = 2^{k+1}$ Below diagonal: $k2^{k-1} \times 2 + 2^k \times 1$ Bottom left: $\frac{k(k-1)}{2}2^{k-2} \times 2 + k2^{k-1} \times 1$	**M1 A1**	**M1** Multiplies the two matrices, attempting at least the entries below the diagonal. Condone slips. **A1** All nine entries correct, unsimplified.
	$k2^k + 2^k = (k+1)2^k$ $\left(\frac{k(k-1)}{2} + k\right)2^{k-1} = \frac{k(k+1)}{2}2^{k-1}$ $= \frac{(k+1)k}{2}2^{(k+1)-2}$	**A1**	cso. Every entry in the $n = k+1$ form, the bottom-left reaching $\frac{k(k+1)}{2}2^{k-1}$ oe by correct algebra.
	True for $n = 1$; true for $n = k \Rightarrow$ true for $n = k + 1$ $\Rightarrow$ true for all $n \geq 1$	**E1**	Concludes with all three: true for $n = 1$; true for $n = k$ implies true for $n = k + 1$; hence true for all integers $n \geq 1$.
	Total for Question 4	**6**	

Question	Scheme	Marks	Notes
5(a)	$n=1$: $\begin{pmatrix}(3c)^1 & 1(3c)^0\\ 0 & (3c)^1\end{pmatrix}=\begin{pmatrix}3c & 1\\ 0 & 3c\end{pmatrix}=\mathbf{M}$	**B1**	Needs the RHS with $n=1$ substituted shown equal to $\mathbf{M}$.
	Assume $\mathbf{M}^k=\begin{pmatrix}(3c)^k & k(3c)^{k-1}\\ 0 & (3c)^k\end{pmatrix}$ $\mathbf{M}^{k+1}=\mathbf{M}^k\mathbf{M}$	**M1**	States the assumption for $n=k$ and writes $\mathbf{M}^{k+1}$ as $\mathbf{M}^k\mathbf{M}$ or $\mathbf{M}\mathbf{M}^k$.
	$\begin{pmatrix}(3c)^k & k(3c)^{k-1}\\ 0 & (3c)^k\end{pmatrix}\begin{pmatrix}3c & 1\\ 0 & 3c\end{pmatrix}$ $=\begin{pmatrix}(3c)^{k+1} & (3c)^k+3ck(3c)^{k-1}\\ 0 & (3c)^{k+1}\end{pmatrix}$	**M1 A1**	**M1** Multiplies the assumed $\mathbf{M}^k$ and $\mathbf{M}$, in either order, by rows and columns. Condone slips in entries. **A1** Correct product, any unsimplified form.
	$=\begin{pmatrix}(3c)^{k+1} & (k+1)(3c)^k\\ 0 & (3c)^{k+1}\end{pmatrix}$	**A1**	Top-right entry reaches $(k+1)(3c)^k$, oe such as $(k+1)(3c)^{(k+1)-1}$, without error.
	True for $n=1$; true for $n=k\Rightarrow$ true for $n=k+1$; so true for all positive integers n	**E1**	States the base case, the implication from $n=k$ to $n=k+1$, and truth for all positive integers n.
		[6]	
5(b)	$\det(\mathbf{M}^n)=(3c)^n(3c)^n-n(3c)^{n-1}\times 0$	**M1**	Uses $ad-bc$ on the matrix from (a), or $(\det\mathbf{M})^n$.
	$=(3c)^{2n}$	**A1**	oe, such as $(9c^2)^n$ or 9^nc^{2n}.
	$(9c^2)^n=144^n\Rightarrow 9c^2=144$	**M1 A1**	**M1** Equates their determinant, of the form $(kc^2)^n$ or $(kc)^{2n}$, to 144^n and removes the power n by roots, bases or logs. **A1** oe, such as $3c=12$ or $c^2=16$.
	$c=4$	**A1**	A0 if $c=-4$ is also given as an answer.
		[5]	
	Total for Question 5	**11**	

Question	Scheme	Marks	Notes
6	$n = 1$: $(-1)^1 = -1$, $-1(-1)^1 = 1$, $\frac{1(0)}{2}(-1)^1 = 0$ RHS = given matrix, so true for $n = 1$	**B1**	Evaluates the RHS at $n = 1$ and states it equals the given matrix. Writing the given matrix alone is not enough.
	Assumes true for $n = k$; writes the $(k + 1)$th power as the kth-power matrix times the given matrix	**M1**	States the assumption for $n = k$ and forms the product of the assumed matrix and the given matrix, in either order.
	Leading diagonal: $-(-1)^k$ Above diagonal: $(-1)^k + k(-1)^k$ Top right: $-k(-1)^k - \frac{k(k-1)}{2}(-1)^k$	**M1 A1**	**M1** Multiplies their assumed matrix by the given matrix, finding all entries or the three distinct non-zero types. Condone sign slips. **A1** All entries correct, need not be simplified.
	$(-1)^{k+1}$, $-(k + 1)(-1)^{k+1}$ $\frac{(k+1)k}{2}(-1)^{k+1}$, zeros below the diagonal	**A1**	cso. All entries shown in the given form with $n = k + 1$; top right as $\frac{(k+1)k}{2}(-1)^{k+1}$ oe.
	True for $n = 1$; true for $n = k \Rightarrow$ true for $n = k + 1$ $\Rightarrow$ true for all $n \in \mathbb{Z}^+$	**E1**	Concludes with all three points: true for $n = 1$, true for $n = k$ implies true for $n = k + 1$, so true for all positive integers n.
	Total for Question 6	**6**	

Question	Scheme	Marks	Notes
7	$F_2 = F_1 + F_0 = 1$ $\Rightarrow \mathbf{M}^1 = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} F_2 & F_1 \\ F_1 & F_0 \end{pmatrix}$	**B1**	Needs $F_2 = 1$ from the recurrence and $\mathbf{M}$ matched entry by entry with F_2, F_1, F_0.
	Assume $\mathbf{M}^k = \begin{pmatrix} F_{k+1} & F_k \\ F_k & F_{k-1} \end{pmatrix}$ $\mathbf{M}^{k+1} = \mathbf{M}^k\mathbf{M}$ $= \begin{pmatrix} F_{k+1} & F_k \\ F_k & F_{k-1} \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$	**M1**	States the result for $n = k$ and writes $\mathbf{M}^{k+1}$ as $\mathbf{M}^k\mathbf{M}$ or $\mathbf{M}\mathbf{M}^k$ with the assumed form substituted.
	$= \begin{pmatrix} F_{k+1} + F_k & F_{k+1} \\ F_k + F_{k-1} & F_k \end{pmatrix}$	**M1 A1**	**M1** Multiplies the two matrices row by column to four entries in terms of F; condone one slip. **A1** Correct product, simplified or not, from either order of multiplication.
	$F_{k+1} + F_k = F_{k+2}$, $F_k + F_{k-1} = F_{k+1}$ $\Rightarrow \mathbf{M}^{k+1} = \begin{pmatrix} F_{k+2} & F_{k+1} \\ F_{k+1} & F_k \end{pmatrix}$	**A1**	cso. Uses the recurrence on both sums to reach the $n = k + 1$ form.
	True for $n = 1$; true for $n = k$ $\Rightarrow$ true for $n = k + 1$; so true for all integers $n \geq 1$	**E1**	States true for $n = 1$, that true for $n = k$ implies true for $n = k + 1$, and hence true for all integers $n \geq 1$.
	Total for Question 7	**6**	

Question	Scheme	Marks	Notes
8	$n = 1$: $2^1 - 1 = 1$, $1 + (1-2)2^0 = 0$, $1 \times 2^0 = 1$, $2^1 = 2 \Rightarrow$ formula gives $\mathbf{A}$	**B1**	Formula evaluated at $n = 1$ and shown equal to $\mathbf{A}$. Not for "true for $n = 1$" alone.
	Assumes the formula true for $n = k$ $\mathbf{A}^{k+1} = \mathbf{A}^k\mathbf{A}$, with the assumed $\mathbf{A}^k$	**M1**	States the assumption for $n = k$ and writes $\mathbf{A}^{k+1}$ as $\mathbf{A}^k\mathbf{A}$ or $\mathbf{A}\mathbf{A}^k$ using it.
	Row 1: $1,\ 1 + 2(2^k - 1)$, $(2^k - 1) + 2\left(1 + (k-2)2^{k-1}\right)$ Row 2: $0,\ 2^{k+1},\ 2^k + 2k\,2^{k-1}$ Row 3: $0,\ 0,\ 2^{k+1}$	**M1 A1**	**M1** Multiplies their $\mathbf{A}^k$ and $\mathbf{A}$ row by column, in either order, attempting every non-zero entry. Condone slips. **A1** Correct product, unsimplified, oe for $\mathbf{A}\mathbf{A}^k$.
	Entries $2^{k+1} - 1$, $1 + (k-1)2^k$, $(k+1)2^k$, 2^{k+1} $\Rightarrow$ the formula with $n = k + 1$	**A1**	cso. Every entry shown to match the formula with $n = k + 1$; $1 + (k-1)2^k$ and $(k+1)2^k$ need not be rewritten in terms of $k + 1$.
	True for $n = 1$; if true for $n = k$, true for $n = k + 1$ $\Rightarrow$ true for all positive integers n	**E1**	Concludes with all three: true for $n = 1$; true for $n = k$ implies true for $n = k + 1$; hence true for all positive integers n.
	Total for Question 8	**6**	